material, depth of cut, feed, type of operation and the coolant used.

Example

Calculate the cutting speed to perform milling with a cutter of diameter 60 mm and spindle speed of 250 rpm.

Solution

Given Diameter of cutter (D) = 60 mm

Spindle speed (N) = 250 rpm

$$\text{Cutting speed} = \frac{\pi dn}{1000} \text{ meter/minute}$$

$$\pi = \frac{22}{7} \text{ (or) } 3.14,$$

$$= \frac{22}{7} \times \frac{60 \times 250}{1000}$$

$$= 47.14 \text{ metre/minute}$$

Feed

The feed in a milling machine is defined as the distance the workpiece advance under the cutter. Feed can be expressed in three different methods.

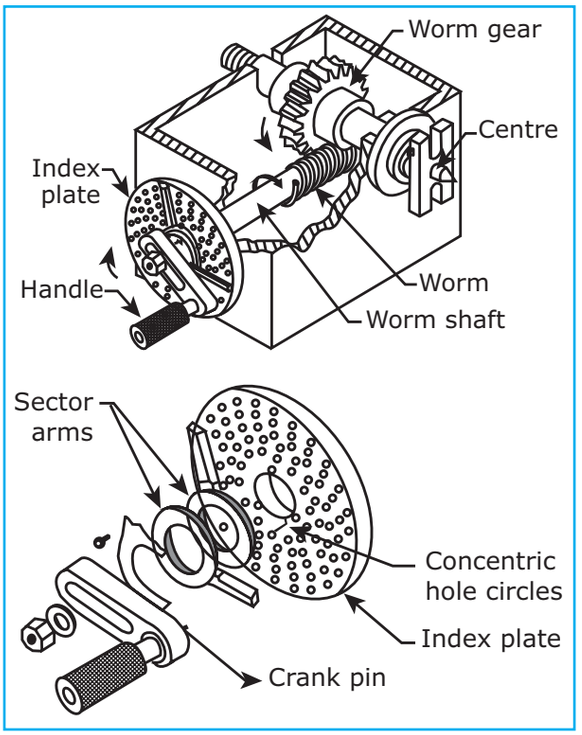
1. **Feed per teeth:** It is the distance the work advances in time between engagements by the two successive teeth. It is expressed.
2. **Feed per cutter revolution:** It is the distance the work advance in the time when the cutter turns through one complete revolution. It is expressed in mm per revolution of the cutter.
3. **Feed per minute:** It is the distance the work advances in one minute. It is expressed in mm per minute. The feed in a milling machine depends on the material to be machined, cutter material, depth of cut, cutting speed, type of operation and the rigidity of the machine.

Depth of cut

The depth of cut is the thickness of the material removed in one pass of the work below the cutter. It is expressed in mm.

5.17 Indexing head

Indexing is the method of dividing the periphery of a piece of work into any number of equal parts. The attachment used for performing indexing is known as indexing head.



Indexing head

Uses of indexing

The indexing operation can be adapted for cutting gears, ratchet wheels. Keyways, fluted drills taps and reamers. The indexing head serves as an attachment for holding and indexing the work in during the above tasks. There are three different types of indexing heads namely

1. Plain or simple dividing head
2. Universal dividing head
3. Optical dividing head

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